Leakage

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for Leakage before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sampling and Intended Use

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Inspect Before Modeling

Without opening the old lesson, explain one key idea from Inspect Before Modeling. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Features, Labels, Targets

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Leakage trace and record the first step you can explain without help.

Use W2: Distinguish frames from **group**s for Splits and related examples. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Leakage explanation against the supplied facts before discussing it.

Use the worked case for Splits and related examples. Proposed explanation: “Different row IDs mean independent examples”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Each of five plants has four photos. A random photo split puts different views of every plant in both **training** and **test set**s. Explain the problem and repair.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Leakage target to a trace and a justified conclusion.

Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected **group**ing rule.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Leakage case and show the deciding evidence.

Ten photos per leaf are randomly split so one leaf appears in train and test.

Explain and repair **leakage**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 26

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A feature records the expert's final species decision, made after the **prediction time**. Why is this **leakage**, and what should happen to it?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 8 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Leakage adds to Responsible Plant Classifier and keep the evidence.

Audit filenames, metadata, post-diagnosis fields, and duplicate plant identities; record every removed **leakage** path.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Five photos of each leaf are randomly split by file into train and test. Why can accuracy be optimistic? Give a corrected **group**ing rule.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Explain the three roles

> Explain the three roles. Apply the supplied Leakage facts and keep your reasoning.

Explain what **training**, validation and **test set**s are used for. What changes if you repeatedly choose a model using its test score?

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Distinguish frames from groups

> Distinguish frames from groups. Apply the supplied Leakage facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Leakage", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your

> workbook while practising.

For the worked A through F plan, give **training**, validation and test counts first in **group**s and then in rows. Interpret [] from the code and state one problem it cannot detect.

> **My reasoning and evidence**

WORKBOOK · 14 OF 26

W3 Repair a related example leak

> Repair a related example leak. Apply the supplied Leakage facts and keep your reasoning.

A1 and A2 are in **training** and A3 is in test. Explain why unique frame IDs and a fixed random seed do not fix this. Propose a repair and what to record.

> **My reasoning and evidence**

WORKBOOK · 15 OF 26

W4 Audit and repair independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Audit and repair independently. Apply the supplied Leakage facts and keep your reasoning.

Eight frames come from trips P (P1 P2 P3), Q (Q1 Q2), R (R1), S (S1 S2). A proposed split is **training** P1 Q1 R1; validation P2 S1; test Q2 P3 S2. Identify shared **group**s for each pair of sets.

Repair it by assigning P and Q to **training**, R to validation and S to test. List all frames in each role and give row and **group** counts. State one limit of the repaired plan.

My reasoning and evidence

W5 Match separation to the claim

> Match separation to the claim. Apply the supplied Leakage facts and keep your reasoning.

A model will be evaluated on new routes. Route North has trips A and B; route South has C and D. A proposed split trains on A and C and tests on B and D. Explain the problem and give a route-based alternative.

Is this enough for a strong performance claim?

My reasoning and evidence

WORKBOOK · 17 OF 26

Scaling without leakage — W1 Learn the scale from training

Use the supplied Leakage task to record your reasoning, evidence and next step.

Using A,B,C, give the minimum, maximum and **range** for each feature. **Transform** all three **training** rows and query (11,120). Explain which rows supplied the scale.

My reasoning and evidence

WORKBOOK · 18 OF 26

Scaling without leakage — W2 Trace the changed prediction

Use the supplied Leakage task to record your reasoning, evidence and next step.

For query (11,120), calculate raw and scaled distances to A,B,C. Give both k=1 **prediction**s.

Does this prove that scaling always improves **prediction**s?

My reasoning and evidence

WORKBOOK · 19 OF 26

Scaling without leakage — W3 Keep one coordinate system

Use the supplied Leakage task to record your reasoning, evidence and next step.

Transform (12,80) using the stored **training** scale. Explain the negative value and why validation must not fit its own scale. Give the worked validation and baseline scores.

My reasoning and evidence

WORKBOOK · 20 OF 26

Scaling without leakage — W4 Scale a fresh dataset independently

Use the supplied Leakage task to record your reasoning, evidence and next step.

New **training** rows are P (20,1000) clear, Q (40,600) blocked, R (40,1000) blocked. Fit minima and **range**s. For query (22,680), show **transform**ed features, raw and scaled distances to P,Q,R, and both k=1 **prediction**s.

Transform new row (50,520) with the same scale.

My reasoning and evidence

WORKBOOK · 21 OF 26

Scaling without leakage — W5 Repair two preprocessing mistakes

> Use the supplied Leakage task to record your reasoning, evidence and next step.

A learner fits scales using **training** plus validation features but no labels. Another divides by a zero **training range**. Explain both problems and give a recorded repair for each.

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Leakage

> Split by plant identity when testing generalisation to new plants. Explain the labelled step before checking the changed condition.

Split by plant identity when testing generalisation to new plants.

Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/8

Split by plant, not by view

	view1	view2	view3
Cedar	train	train	train
Elm	train	train	train
Fir	test	test	test

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/8

Leakage

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Group related views

All three Cedar views stay together; likewise Elm and Fir.

↓

2. Assign groups

Training receives six views from Cedar/Elm; test receives three Fir views.

↓

3. State claim

The final test probes one unseen plant, not three independent plants.

CHANGE ONE THING

Move Fir's first view into training while retaining its other two in test.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

WORKBOOK · 24 OF 26

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Split by plant identity when testing generalisation to new plants.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The final test probes one unseen plant, not three independent plants.

1. Assign **group**s

2. State claim

3. **Group** related views

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: State claim The final test probes one unseen plant, not three independent plants. Read the picture in order.

The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move Fir's first view into **training** while retaining its other two in test. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. The unseen-plant evaluation is compromised.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The unseen-plant evaluation is compromised. Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

Repair a missing building block

Review Sampling and Intended Use (ai-4/7). Goal: Inspect row meaning missing values and repeated records. Short practice: Which collection would address the intended-use gap most directly?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Inspect Before Modeling (ai-4/5). Goal: Inspect row meaning missing values and repeated records. Short practice: Which original information should the inspection log preserve?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Features, Labels, Targets (ai-4/2). Goal: Separate **prediction input**s from observed targets. Short practice: Why can a historical table contain a field that is an invalid **input**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

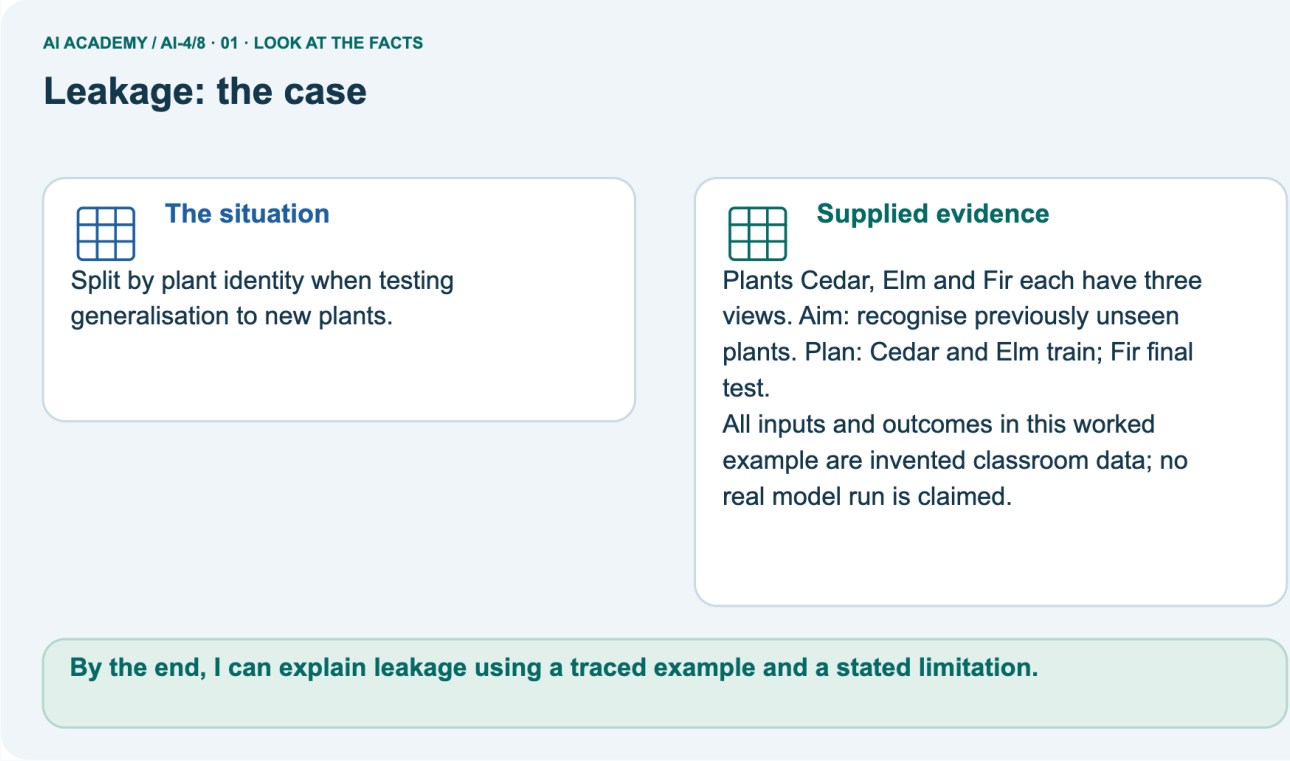
What I repaired, the evidence I kept and the next test

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Leakage picture from its labels, then record a relevant limit.

01. Leakage: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

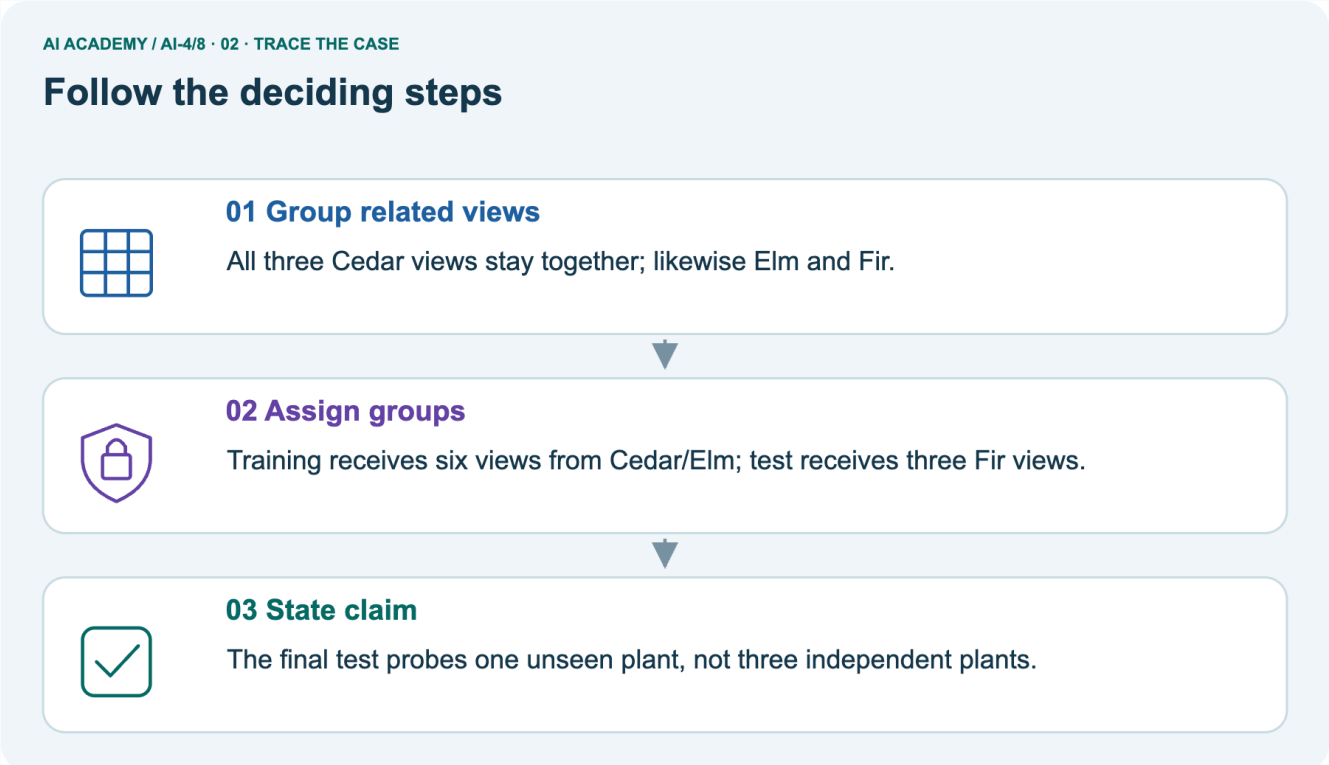
Split by plant identity when testing generalisation to new plants. Supplied facts: Plants Cedar, Elm and Fir each have three views. Aim: recognise previously unseen plants. Plan: Cedar and Elm train; Fir final test.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



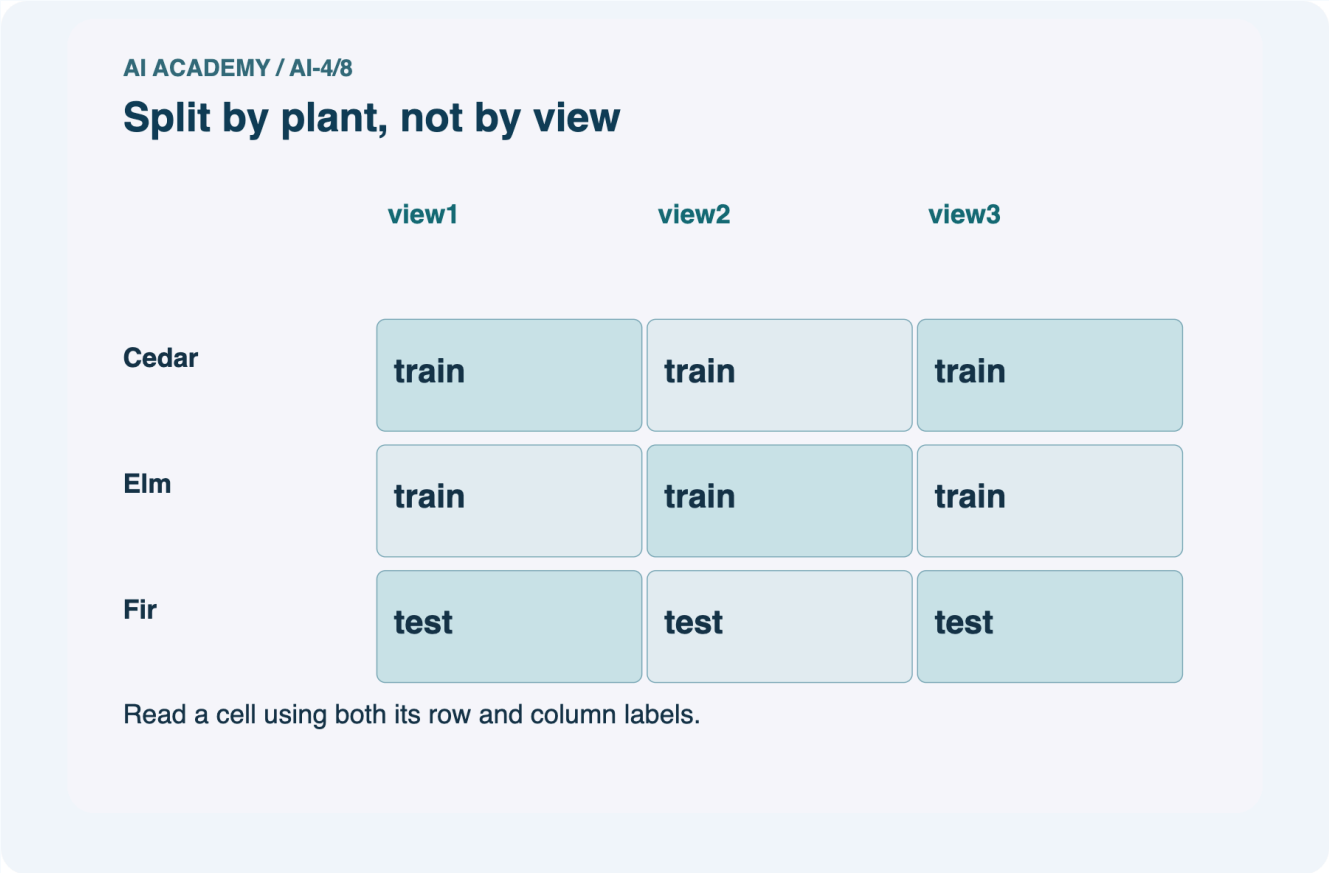
Follow the labelled picture with the matching explanation. Open the vector diagram

Group related views: All three Cedar views stay together; likewise Elm and Fir. → Assign **group**s: **Training** receives six views from Cedar/Elm; test receives three Fir views. → State claim: The final test probes one unseen plant, not three independent plants.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Split by plant, not by view



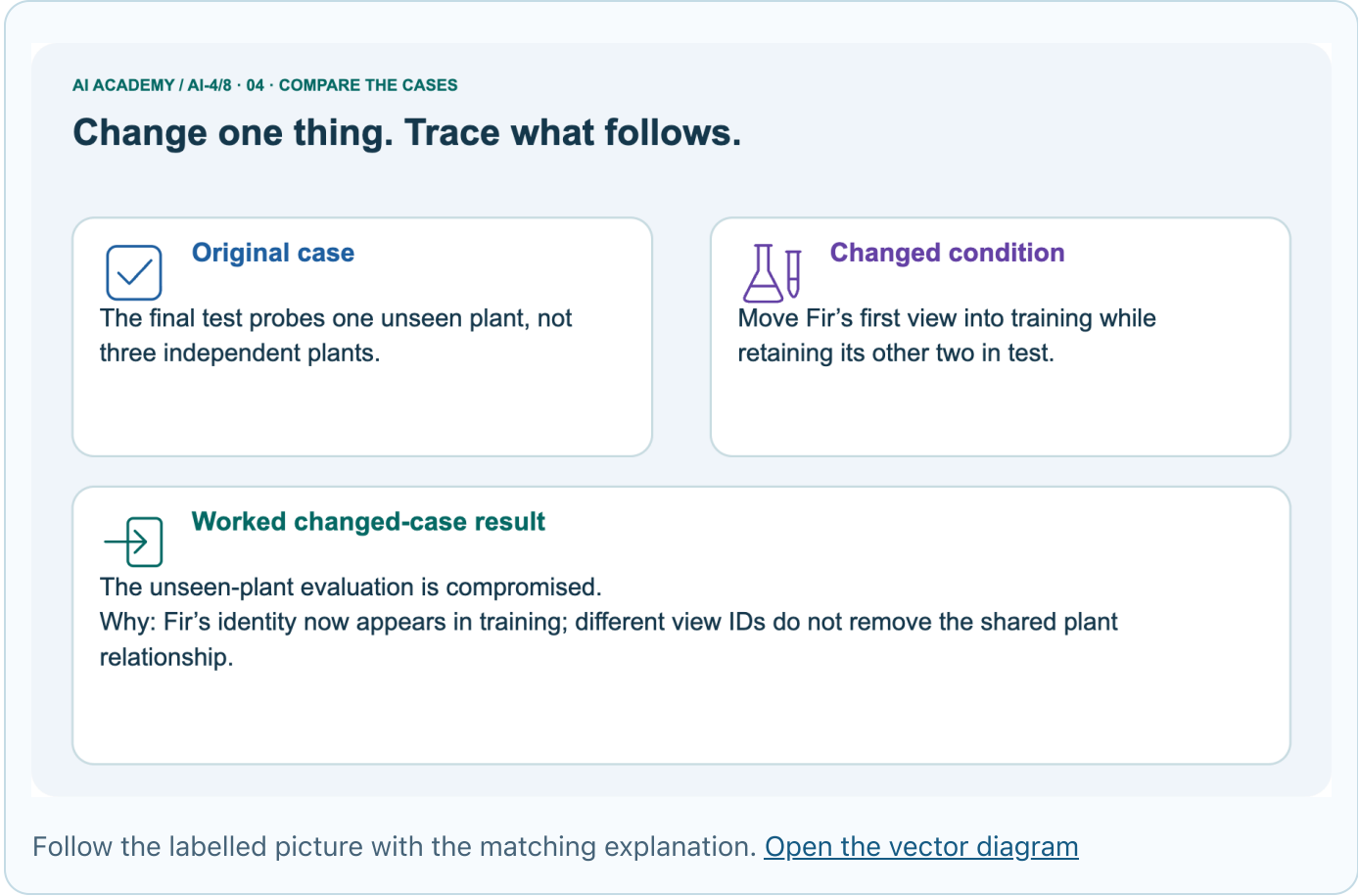
Follow the labelled picture with the matching explanation. Open the vector diagram

Split by plant, not by view. Cedar: view1 = train, view2 = train, view3 = train. Elm: view1 = train, view2 = train, view3 = train. Fir: view1 = test, view2 = test, view3 = test.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

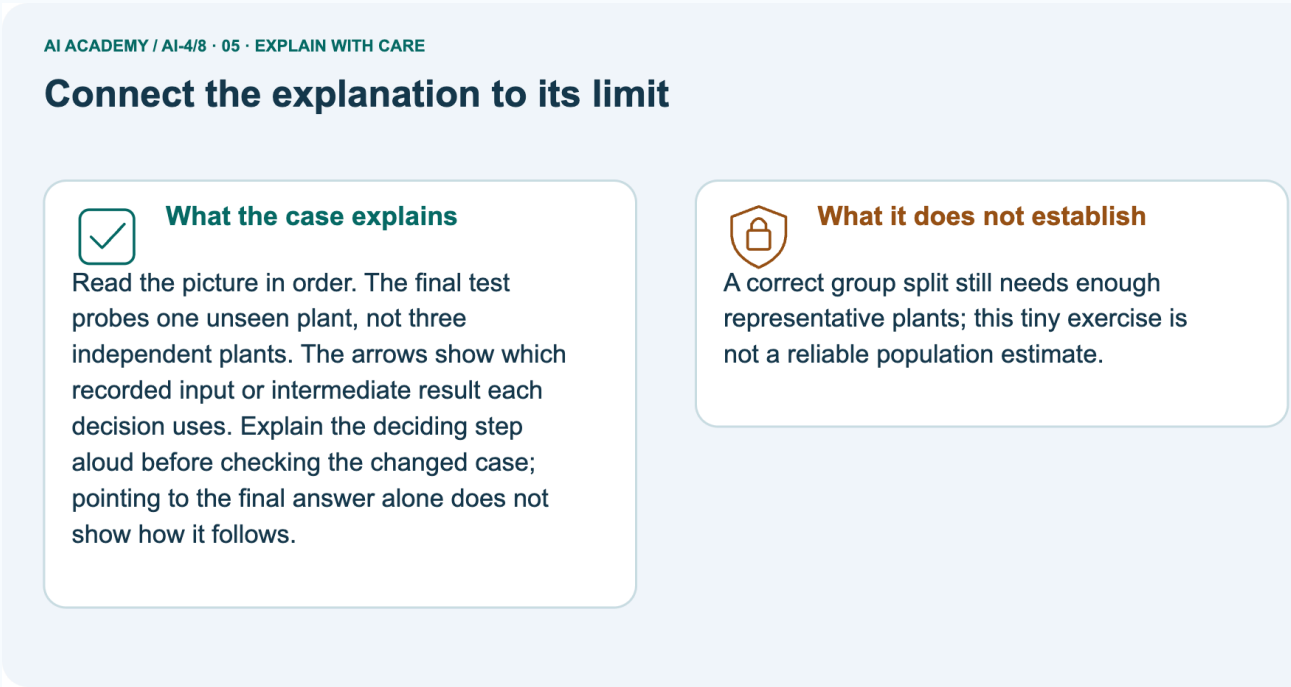
Original case: The final test probes one unseen plant, not three independent plants. Changed condition: Move Fir's first view into **training** while retaining its other two in test.

Expected result: The unseen-plant evaluation is compromised. Reason: Fir's identity now appears in **training**; different view IDs do not remove the shared plant relationship.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. The final test probes one unseen plant, not three independent plants. The arrows show which recorded **input** or intermediate result each decision uses.

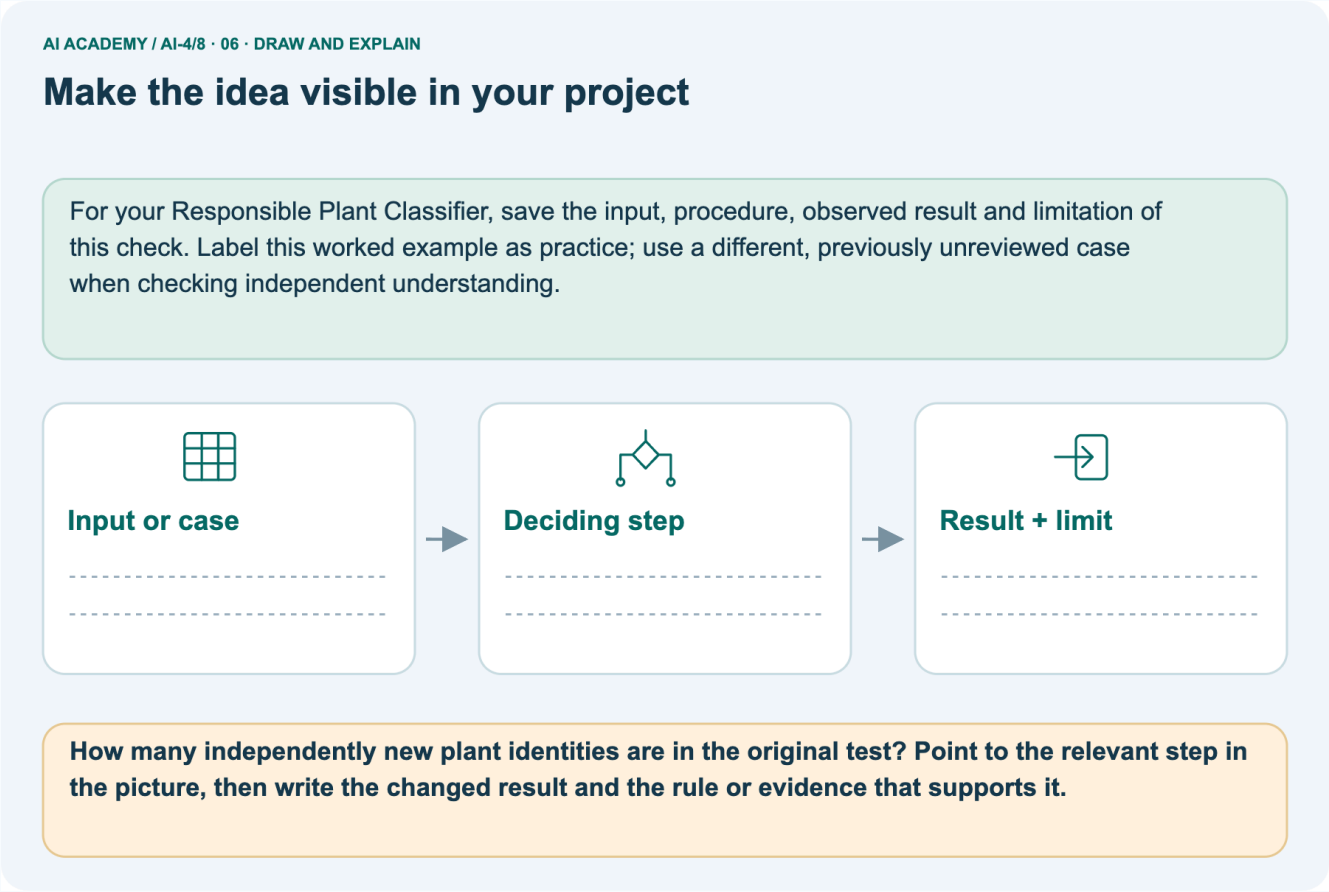
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A correct **group** split still needs enough representative plants; this tiny exercise is not a reliable population estimate.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many independently new plant identities are in the original test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.